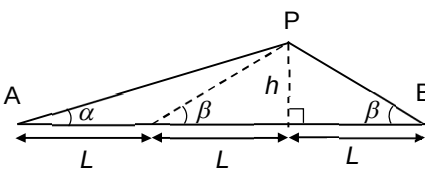
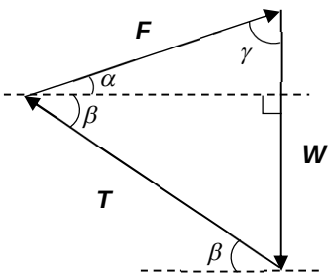


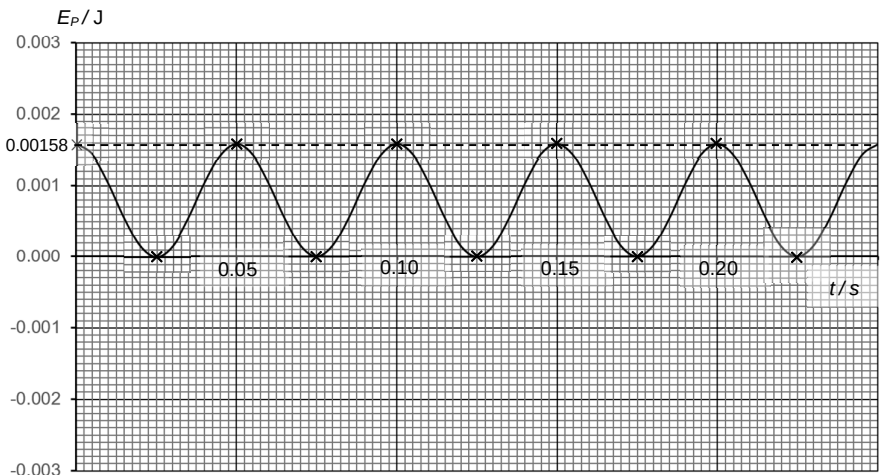
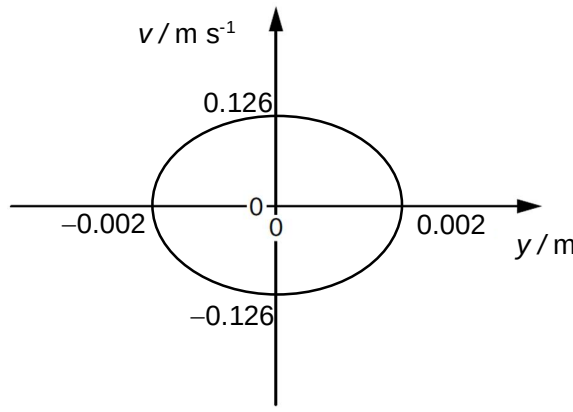
30	Answer: C
	Number of reactions = $0.0010 / 1.008u = 5.976 \times 10^{23}$ Energy released in one reaction $= [7.018u + 1.008u - 2(4.004u)]c^2$ $= 2.689 \times 10^{-12} \text{ J}$ Total energy released $= 2.689 \times 10^{-12} \times 5.976 \times 10^{23}$ $= 1.61 \times 10^{12} \text{ J}$

Paper 2 – Structured Questions

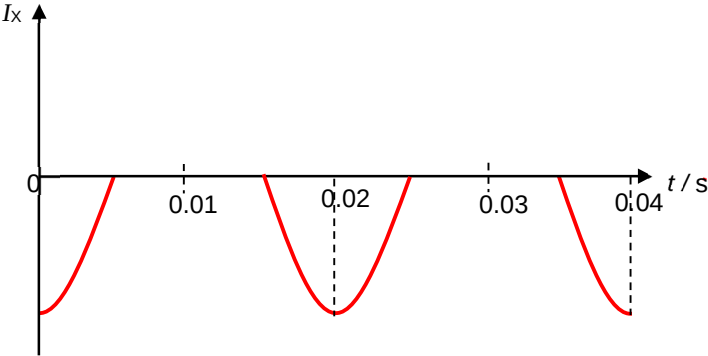
Qns	Answer	Marks
1(a)	Resultant force acting on the body is zero. Resultant moment OR Torque <u>about any point</u> is zero.	B1 B1
1(b)	Point at which <u>whole weight of body may be considered / seems to act.</u>	B1
1(c)(i)	Since T and W have zero moment about P (because their line of actions passes through P), so F must have zero moment, i.e. line of action of F must pass through P OR When all forces have lines of actions that pass through P, (perpendicular) distance from P is zero for all forces (M1), so sum of moments about P is zero	B1 B1

1(c)(ii)1.	By principle of moment and taking moments about A, $2W = 3T \sin \beta$ $2(10) = 3T \sin 30^\circ$ $T = 13.3 \text{ N}$	B1 A1
1(c)(ii)2.	Sum of forces in the vertical direction and horizontal direction are both zero. Considering horizontal forces: $F \cos \alpha = T \cos \beta \quad (1)$ Considering vertical forces: $W = F \sin \alpha + T \sin \beta \Rightarrow F \sin \alpha = W - T \sin \beta \quad (2)$ $\frac{\text{Eq(2)}}{\text{Eq(1)}}: \tan \alpha = \frac{W - T \sin \beta}{T \cos \beta} = \frac{10 - 13.3 \sin 30}{13.3 \cos 30}$ $\alpha = 16.2^\circ$ (Physics statements to be present to score full credit) <u>Alternative method (for both parts):</u>  $\tan \alpha = \frac{h}{2L}$ $\tan \beta = \frac{h}{L}$ $\tan \alpha = 0.5 \tan \beta$ $= 0.5 \tan 30^\circ$ $\alpha = 16^\circ$  $\gamma = 90^\circ - \alpha = 90^\circ - 16^\circ = 74^\circ$ Using sine rule, $\frac{T}{\sin \gamma} = \frac{W}{\sin(\alpha + \beta)}$ $T = \frac{10}{\sin(30^\circ + 16^\circ)} \times \sin 74^\circ$ $= 13.4 \text{ N}$	B1 Horizontal B1 Vertical A1

Qns	Answer	Marks
2(a)(i)	An arrow originating from the object, pointing towards point O, and labelled Z.	B1
2(a)(ii)	The frictional force provides the centripetal force, and hence must point towards the centre of the disc.	B1
2(b)(i)	Using points (6.00, 5.90) and (9.00, 8.90) on the line Gradient = $\frac{8.90 - 5.90}{9.00 - 6.00} = 1.00$	B1 B1
2(b)(ii)	$\text{gradient} = \frac{\omega_{\max}^2}{\frac{1}{r}} = r \omega_{\max}^2 = a_{c(\max)}$ Hence, the gradient is numerically equal to the <u>maximum centripetal acceleration</u> .	B1 A1*
2(c)	$f = \text{gradient} \times m$ $= 1.00 \times 0.80$ $= 0.80 \text{ N}$	C1 A1
(d)	As angular speed increases, required centripetal force increases. (friction provides required centripetal force) When required centripetal force exceeds maximum frictional force possible, object slides.	B1 B1

Qns	Answer	Marks
3(a)(i)	Free oscillations are oscillations with constant amplitude and <u>without</u> energy loss or gain as there is no driving or resistive forces acting on it.	B1
3(a)(ii)	Natural frequency is the frequency at which a body will vibrate when there is <u>no driving</u> or <u>no resistive forces</u> acting on it.	B1
3(b)(i)	 $E_{p(max)} = \frac{1}{2} m y^2 \omega^2 = \frac{1}{2} (0.200) (0.002)^2 \left(\frac{2\pi}{0.1} \right)^2$ $= 0.00158 \text{ J}$	B1 correct sine-square shape, all above x-axis B1 correct period & phase B1 correct max E_p indicated
3(b)(ii)	 $v_0 = \omega y_0 = \frac{2\pi}{T} y_0 = \frac{2\pi}{0.1} (0.002) = 0.126 \text{ m s}^{-1}$	B1 Correct & smooth shape B1 correct max v caln at y=0 B1 correct max y at v=0
3(b)(iii)	Period decrease. Equilibrium point is raised once the mass is removed. Hence amplitude increases.	B1 B1

Qns	Answer	Marks
4 (a)(i)	The Principle of Conservation of Linear Momentum states that the total linear momentum of an isolated system of interacting bodies before and after collision remains constant if no net external force acts on the system. OR The Principle of Conservation of Linear Momentum states that <u>the total linear momentum of a system remains constant provided that no resultant external force acts on the system</u>.	B1
(a)(ii)	The resultant force acting on a body is proportional to the rate of change of momentum of that body.	B1
(b)(i)	After collision, the bodies will move along the same line that joins the 2 bodies' centre of mass rightwards.	B1
(b)(ii)	By principle of conservation of momentum: $mu + 14m(0) = 15mV$ $V = \frac{1}{15}u$ Ratio of kinetic energy $= \frac{KE_{\text{nitrogen}}}{KE_{\text{neutron}}} = \frac{\frac{1}{2}MV^2}{\frac{1}{2}mv^2} = \frac{15\cancel{m} \left(\frac{\cancel{u}}{15} \right)^2}{\cancel{m} \cancel{u}^2} = \frac{1}{15} = 0.067$	C1 M1 A1
(b)(iii)	From (bii), since initial total KE before collision not equals to final total KE after collision, the collision is not elastic (inelastic). OR Since there is no relative separation between the neutron and nitrogen atom after collision, the relative speed of approach is not equals to relative speed of separation, the collision is not elastic (inelastic).	M1 A1

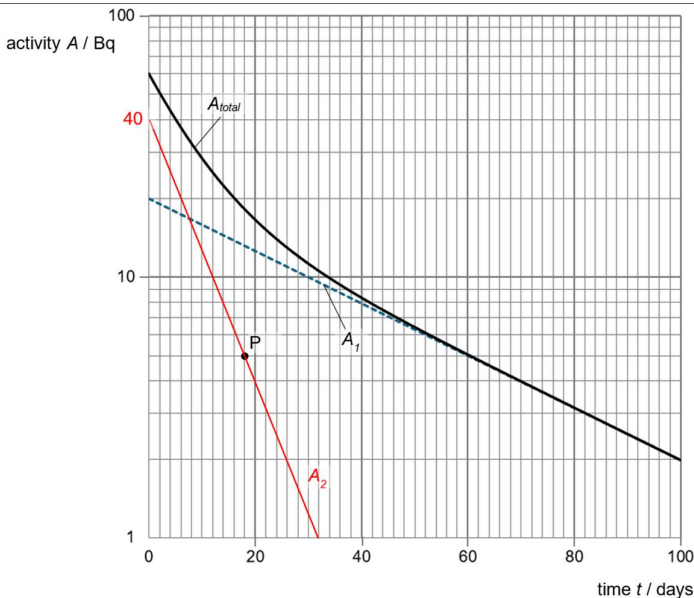
Qns	Answer	Marks
5 (a)	When S is opened, current in coil Y stops flowing and the magnetic flux produced by the current in coil Y decreases. Hence <u>magnetic flux density through coil X decreases.</u> By Faraday's Law, an <u>electromotive force</u> is <u>induced in coil X.</u> By Lenz's Law, the induced emf in coil X would be <u>in a direction</u> so as to produce effects to <u>oppose the decrease in the magnetic flux density.</u> The induced emf would have produce current, in coil X in the <u>clockwise direction when viewed from Q.</u> But since this current would be in <u>reversed biased direction, LED remain unlit.</u> (For last B1, Student to either describe direction of current to be CW (from Q) leading to LED unlit OR describe current being in reversed biased direction leading to LED unlit.)	B1 B1 B1 B1
5b		B1 Negative cosine graph B1 Rectified in the correct regions

Qns	Answer	Marks
6(ai)	Lamination increases resistance, thus <u>reduces induced (eddy) current in the iron core</u> , which would reduce heat loss.	M1
6(aii)	Alternating current in primary coil (I_{pri}) give rise to changing magnetic flux (ϕ) in the primary coil and the iron core. Thus primary coil current is in phase with the magnetic flux. This changing magnetic flux (ϕ) links the secondary coil through the iron core. E.m.f. induced in secondary coil (E_{sec}) is proportional to <u>rate</u> of change of flux (ϕ). Secondary coil e.m.f. (E_{sec}) is not in phase with the flux (ϕ). (Hence e.m.f induced in secondary coil not in phase with alternating current in primary coil.	B1 B1 B1
6(b)	$\left(\frac{V_{sec}}{V_{pri}}\right)_{rms} = \frac{N_{sec}}{N_{pri}}$ $(V_{sec})_{rms} = \frac{8100}{300} \times 9.0 = 243 \text{ V}$ $\text{Peak voltage across load} = (V_{sec})_{rms} \times \sqrt{2} = 243 \times \sqrt{2}$ $= 3.4 \times 10^2 \text{ V}$	C1 A1

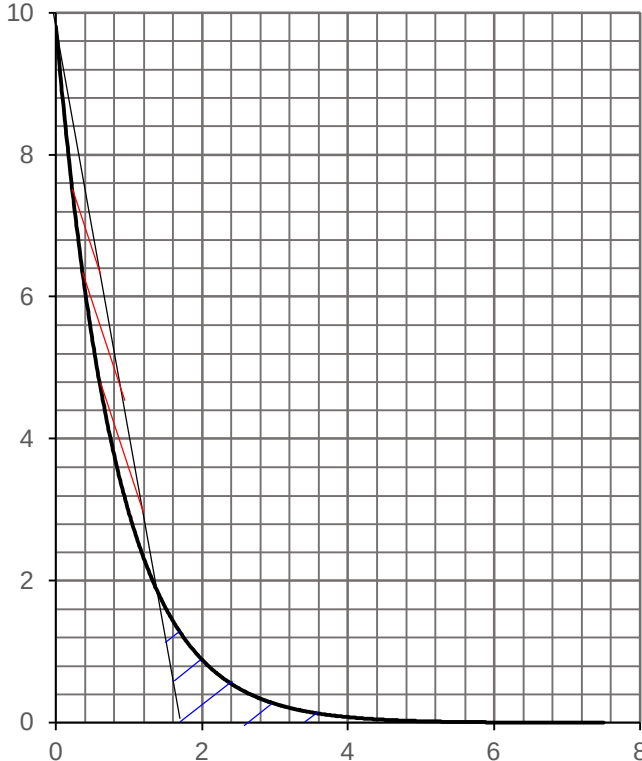
Qns	Answer	Marks
7(a)(i)	The force on the electron is (always) perpendicular to the <u>velocity/</u> perpendicular to the <u>direction of travel/motion</u> of the electron. Thus no work is done by the force on the particle. (The speed remains unchanged and the force changes its direction.)	B1
7(a)(ii)	$F = Bqv$ $= (0.0050)(1.60 \times 10^{-19})(5.0 \times 10^6)$ $= 4.0 \times 10^{-15} \text{ N}$	C1 A1
7(a)(iii)	Magnetic force provides the centripetal force. $F = ma$ $= m \left(\frac{v^2}{r} \right)$ $r = \frac{mv^2}{F}$ $= \frac{(9.11 \times 10^{-31})(5.0 \times 10^6)^2}{4.0 \times 10^{-15}}$ $= 5.69 \times 10^{-3}$ $= 5.7 \text{ mm}$ $F = ma$ $(Bqv) = m \left(\frac{v^2}{r} \right)$ $r = \frac{mv}{Bq}$ $= \frac{(9.11 \times 10^{-31})(5.0 \times 10^6)}{(0.0050)(1.60 \times 10^{-19})}$ $= 5.69 \times 10^{-3}$ $= 5.7 \text{ mm}$	B1 B1 c
7(a)(iv)	Radius of path is proportional to its mass. Mass of proton is <u>many order of magnitude larger</u> than the electron implies the radius of the proton would be many orders of magnitude larger (and is too large as compared to a typical laboratory).	B1 B1
7(b)	$F = ma \Rightarrow eE = ma \Rightarrow e \frac{V}{d} = ma \Rightarrow a = \frac{eV}{md} \quad \dots\dots (1)$ $t = \frac{x}{v} \quad \dots\dots (2)$ Substituting (1) & (2): $y = \cancel{vt} + \frac{1}{2}at^2$ $= \frac{1}{2} \left(\frac{eV}{md} \right) \left(\frac{x}{v} \right)^2$ $= \left(\frac{eV}{2mdv^2} \right) x^2$	M1 M1 Correct substitution into the correct kinematics formula A1

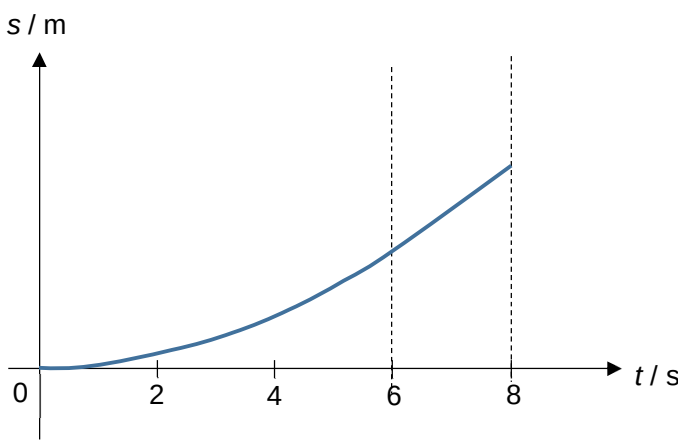
Qns	Answer	Marks
8(a)(i)1	α particles have <u>greater ionisation power</u> than X-rays, γ rays and energetic β rays. Hence, there is <u>greater damage</u> (biological effect stated in the last sentence in the 2nd last para on pg 18) produced by α particles.	B1 B1
8(a)(i)2	Neutrons can be absorbed by the nucleus of an atom to <u>form a radioactive isotope</u> , which is unstable and produces secondary <u>ionising radiations</u> .	B1
8(a)(ii)	Shield from source using suitable materials. Increase the distance from a source.	B2
8(a)(iii)	Absorbed dose $= \frac{\text{energy deposited per year}}{\text{mass of tissue}}$ $= \frac{(\text{no. of decay})(\text{energy released per decay})}{\text{mass of tissue}}$ $= \frac{[(3.70 \times 10^4)(365 \times 24 \times 3600)][(5.23 \times 10^6)(1.60 \times 10^{-19})]}{2.00}$ = 0.489 Gy Dose equivalent $= (\text{absorbed dose}) \times \text{RBE}$ $= 0.489 \times 20$ $= 9.8 \text{ Sv year}^{-1}$	C1 C1 A1

8(b)(i)	<u>Gamma rays can pass through the tissue of the body</u> to be captured by the camera outside while <u>beta particles will be stopped</u> by the body tissue.	A1
8(b)(ii)	$\lambda_E = \lambda_B + \lambda_T$ $\frac{\ln 2}{t_E} = \frac{\ln 2}{t_B} + \frac{\ln 2}{t_T}$ $t_E = \frac{t_T t_B}{t_T + t_B}$	B1
8(b)(iii)1	$A_0 = \lambda N_0$ $= \left(\frac{\ln 2}{t_T} \right) \left(\frac{M}{\text{molar mass}} \times N_A \right)$ $= \left(\frac{\ln 2}{6.02 \times 3600} \right) \left(\frac{1.0 \times 10^{-12}}{99} \times 6.02 \times 10^{23} \right)$ $= 1.94 \times 10^5$ $= 1.9 \times 10^5 \text{ Bq}$	B1 B1
8(b)(iii)2	$t_E = \frac{t_T t_B}{t_T + t_B}$ $= \frac{6.02 \times 24}{6.02 + 24}$ $= 4.8 \text{ h}$	A1
8(b)(iii)3	$A = \left(\frac{1}{2} \right)^{t/t_E} A_0$ $= \left(\frac{1}{2} \right)^{\frac{3.0 \times 24}{4.8}} (1.9 \times 10^5)$ $= 5.8 \text{ Bq}$	C1 A1

8(c)(i)	After a long enough time, the activity of the nucleus having the shorter half-life would have decreased to a negligible value compared to the activity of the nucleus having the longer half-life. Hence the total activity would be the same as the activity of the nucleus having the longer half-life.	A1
8(c)(ii)	From graph, time taken for activity of nucleus 1 to decrease from an initial value of 20 Bq to 10 Bq is 30 days, 10 Bq to 5 Bq is 30 days. Hence average half-life = 30 days	A1
8(c)(iii)	At $t = 0$ day, $A_{total} = A_1 + A_2$ $60 = 20 + A_2$ $A_2 = 40 \text{ Bq}$	A1
8(c)(iv)		B1 - line passes through point P and y-intercept at 40 Bq - line must be labelled
8(c)(v)	From graph, time taken for activity of nucleus 2 to decrease from an initial value of 40 Bq to 20 Bq is 6 days, 20 Bq to 10 Bq is 6 days, 10 Bq to 5 Bq is 6 days. Hence average half-life is 6 days.	B1

9749/J2H2PRELIM/2024

Qns	Answer	Marks
2(a)	Acceleration is the rate of change of velocity.	B1
(b)(i)	As the object accelerates downwards, its velocity increases with time. Since <u>the drag force (due to viscous fluid) on the object increases with speed</u> , <u>net force</u> (weight – resistive/drag force) acting on the object <u>decreases</u> , hence the acceleration decreases with time.	B1 B1
(b)(ii)	Initially the velocity of the object is zero, hence the <u>drag force is also zero</u> . The <u>acceleration of the object is only due to its weight / gravitational force</u> , hence acceleration 9.8 m s^{-2} . (Assume upthrust is negligible since given that acceleration is 9.8 m s^{-2})	B1
(b)(iii)	Change in velocity = area under $a-t$ graph $v - 0 \approx (50 \text{ small squares})(0.4 \times 0.4) = 8.0 \text{ m s}^{-1}$ (range: $7.5\text{-}9.0 \text{ m s}^{-1}$) One way to estimate the area under the curve is to draw a line as shown in the diagram below. The line chosen is such that the area shaded in red is roughly equal to the area shaded in blue. The area under the original curve is then roughly equal to the area of the triangle, with base 1.7 and height 9.8. Thus, velocity = area under curve $\approx \frac{1}{2} \times 9.8 \times 1.7 = 8.3 \text{ m s}^{-1}$. Better estimation of the area can be achieved by using more triangles/trapeziums to approximate it.  Some students used kinematics equations without understanding that the acceleration graph already shows that the acceleration is not constant.	C1 A1

(b)(iv)	 Gradient = 0 at $t = 0$ s, then gradually increasing till $t \approx 5.2$ s . After $t \approx 6$ s, gradient is constant (terminal velocity reached) at the value that was reached at 6 s - no kink allowed.	B1 B1
Qns	Answer	Marks
3(a)(i)	$R_{total} = R_{//} + r$ $= \left(\frac{1}{600} + \frac{1}{3000} \right)^{-1} + 30 = 530 \, \Omega$ $V_{LDR} = V_{terminal} = \varepsilon - Ir$ $= 12 - \left(\frac{12}{530} \right)(30) = 11.3 \, \text{V}$ $I_{LDR} = \frac{V_{LDR}}{R_{LDR}}$ $= \frac{11.3}{3000} = 3.77 \, \text{mA} = 3.8 \, \text{mA} \, (2\text{sf}) \, (\text{shown})$ OR $R_{//} = \left(\frac{1}{600} + \frac{1}{3000} \right)^{-1} = 500 \, \Omega$ $V_{LDR} = V_{//} = \frac{R_{//}}{R_{//} + r} \times \varepsilon$ $= \frac{500}{500 + 30} \times 12 = 11.3 \, \text{V}$ $I_{LDR} = \frac{V_{LDR}}{R_{LDR}}$ $= \frac{11.3}{3000} = 3.77 \, \text{mA} = 3.8 \, \text{mA} \, (2\text{sf}) \, (\text{shown})$ OR $R_{total} = R_{//} + r$ $= \left(\frac{1}{600} + \frac{1}{3000} \right)^{-1} + 30 = 530 \, \Omega$	M1 M1 M1

	$I_{Tot} = \frac{\varepsilon}{R_{Tot}} = \frac{12}{530} = 0.0226 \text{ A}$ $I_{LDR} = \frac{600}{600 + 3000} \times I_{Tot}$ $= \frac{600}{600 + 3000} \times 0.0226 = 3.77 \text{ mA} = 3.8 \text{ mA (2sf) (shown)}$	
--	---	--

(a)(ii)	$P_{\text{LDR}} = I_{\text{LDR}}^2 R_{\text{LDR}}$ $= (3.8 \times 10^{-3})^2 (3000)$ $= 0.043 \text{ W (2sf)}$	A1
(b)	<u>Explanation 1 (preferred):</u> Resistance of LDR decreases in bright light, thus effective total resistance of entire circuit decreases. Total current I_{Tot} in circuit increases, resulting in larger potential drop V_r across the internal resistance r Since $V_{\text{terminal}} = \varepsilon - V_r = \varepsilon - I_{\text{Tot}} r$, where V_{terminal} is the terminal p.d., ε is the e.m.f. of the cell; thus the terminal p.d. decreases. OR <u>Explanation 2:</u> Resistance of LDR decreases in bright light, thus resistance of the combination of parallel resistors decreases. By potential divider rule, potential difference (p.d.) across parallel resistors decreases. Since terminal p.d. is the same as p.d. across parallel resistor, terminal p.d. decreases.	B1 B1 B1